AI PORTFOLIO PROJECT

Natural Language to PostgreSQL

An Intelligent LangChain Agent for Conversational Database Querying

LangChain · OpenAI GPT · PostgreSQL · RAG · Agentic AI · Python

THE PROBLEM & SOLUTION

⚠ THE PROBLEM

SQL is a barrier.

Business analysts, product managers, and executives need data insights — but cannot write SQL queries.

They rely on engineering teams, creating delays and bottlenecks in decision-making pipelines.

✓ THE SOLUTION

Conversational DB Intelligence.

A LangChain-powered AI Agent translates plain English questions into precise SQL queries and returns human-readable answers — in real time.

Zero SQL knowledge required.

HOW IT WORKS — SYSTEM ARCHITECTURE

01

NL

User Query

Plain English question input

->

02

AI

LLM Agent

OpenAI GPT interprets intent

->

03

SQL

SQL Tool

LangChain SQLDatabaseToolkit generates query

->

04

DB

PostgreSQL

Query executed on real database

->

05

ANS

Answer

Natural language result returned

RAG-Enhanced: The agent retrieves schema metadata from a vector store to improve SQL generation accuracy before querying the live database.

WHY RAG + AGENTIC AI MATTERS

RAG — Retrieval-Augmented Generation

01 Embeds database schema into a vector store

02 Retrieves relevant table/column context before generating SQL

03 Reduces hallucinations — the LLM 'knows' your schema

04 Adapts to any database without retraining the model

Agentic AI — LangChain Agent

01 Multi-step reasoning: plan → tool call → observe → answer

02 Uses tools (SQLDatabaseToolkit) to introspect live schema

03 Self-corrects when SQL fails — retries with better query

04 Extends easily: add web search, charts, or APIs as tools

REAL-WORLD USE CASES

BI

Business Intelligence

Analysts ask "Show me Q3 revenue by region" — no SQL, instant insight.

HC

Healthcare Analytics

Clinicians query patient records: "List diabetic patients over 60 admitted last month."

EC

E-Commerce Ops

Product teams explore sales data: "Which SKUs have < 10 units in stock?"

FN

Finance & Audit

Auditors interrogate transaction tables without engineer support.

CB

Chatbot Backend

Power customer-facing bots that answer queries from live product or order databases.

ED

Education Platforms

Students explore datasets using natural language — learn data skills without SQL friction.

TECH STACK & IMPACT

Technology Stack

- LangChain
- OpenAI GPT
- PostgreSQL
- SQLAlchemy
- Python 3.11
- Vector Store (RAG)
- SQLDatabaseToolkit
- Streamlit / CLI

Key Outcomes

~0 sec — Time to first SQL result via NL query

100% — SQL-free for end users — no training needed

∞ — Scales to any PostgreSQL schema dynamically

RAG — Schema-aware queries reduce hallucination rate

github.com/Sajalagarwal-ca/langchain_query_NL_PostgreSQL_DB